Question	Answer	Marks
6(a)(i)	work done per unit charge	**B1**
	work done moving positive charge from infinity (to the point)	**B1**
6(a)(ii)	field strength = potential gradient	**M1**
	'–' sign included **or** directions discussed	**A1**
6(b)(i)	gain in kinetic energy (= loss in potential energy) = charge × p.d. **or** $qV = ½mv^2$	**M1**
	so v is independent of separation (because separation not in expressions)	**A1**

Question	Answer	Marks
6(b)(ii)	(at x = 0.40 cm), potential = (–) 75 × 0.40 / 1.2 (= (–) 25 V)	C1
	½mv^2 = qV ½ × 4 × 1.66 × 10^{-27} × v^2 = 2 × 1.60 × 10^{-19} × 25	C1
	or	
	a = Vq/dm **and** v^2 = 2as	(C1)
	v^2 = (2 × 75 × 2 × 1.60 × 10^{-19} × 0.40 × 10^{-2}) / (1.2 × 10^{-2} × 4 × 1.66 × 10^{-27})	(C1)
	v = 4.9 × 10^4 m s^{-1}	A1

Question	Answer	Marks
7(a)(i)	gain is constant	M1